

# OGDEN-HINCKLEY, UT, USA

WMO: 725750

Lat: 41.196N

Lon: 112.011W

Elev: 1363

StdP: 85.99

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 24126

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.7      | -10.8 | -17.6                         | 0.9 | -10.1 | -15.6 | 1.1 | -8.4 | 11.0                  | 4.5  | 9.0 | 4.3  | 2.4                   | 170  | 0.544 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.9                   | 35.5            | 16.3 | 34.0 | 15.9 | 32.7 | 15.6 | 18.4                | 29.9 | 17.7 | 29.6 | 17.0 | 29.2 | 3.1                  | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 14.7                            | 12.3 | 22.0 | 13.0 | 11.1 | 22.7 | 12.1 | 10.4 | 22.7 | 57.2          | 29.8 | 54.8 | 29.4 | 52.6 | 29.3 | 20.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       | DB   | WB  |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4)               | 9.6  | 8.2 | 7.0                        | -15.3 | 37.8               | 2.7 | 1.4                                                | -17.2 | 38.8       | -18.8 | 39.6       | -20.4 | 40.4       | -22.4 | 41.4 | (4) |
| (5)               |      |     |                            | -16.1 | 19.4               | 2.6 | 0.6                                                | -18.0 | 19.9       | -19.5 | 20.2       | -21.0 | 20.6       | -22.9 | 21.1 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.2   | -2.0  | 0.8   | 6.2   | 9.7   | 14.7  | 20.5  | 26.0  | 24.3  | 18.8  | 11.1  | 4.6   | -1.3 | (6)  |
| (7)  |                                                                                  | DBStd       | 10.30  | 4.60  | 4.51  | 4.23  | 4.28  | 4.70  | 4.60  | 2.45  | 2.71  | 4.29  | 4.42  | 4.66  | 4.61 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1391   | 372   | 257   | 133   | 57    | 12    | 1     | 0     | 2     | 39    | 169   | 350   | (8)  |      |
| (9)  |                                                                                  | HDD18.3     | 3213   | 630   | 490   | 377   | 260   | 133   | 31    | 0     | 2     | 46    | 225   | 412   | 608  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1822   | 0     | 1     | 14    | 48    | 156   | 316   | 497   | 444   | 265   | 74    | 7     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 604    | 0     | 0     | 0     | 1     | 19    | 97    | 238   | 188   | 58    | 2     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 7199   | 0     | 0     | 1     | 18    | 249   | 1236  | 2953  | 2096  | 617   | 29    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 3353   | 0     | 0     | 0     | 2     | 69    | 545   | 1564  | 976   | 193   | 3     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.1    | 2.3   | 2.8   | 3.3   | 3.6   | 3.3   | 3.4   | 3.2   | 3.2   | 3.0   | 2.7   | 2.6   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 435    | 33    | 34    | 43    | 53    | 48    | 33    | 15    | 19    | 39    | 41    | 40    | 36   | (15) |
| (16) |                                                                                  | PrecMax     | 872    | 80    | 82    | 130   | 143   | 139   | 107   | 48    | 82    | 140   | 107   | 90    | 136  | (16) |
| (17) |                                                                                  | PrecMin     | 210    | 0     | 2     | 2     | 6     | 0     | 0     | 0     | 1     | 0     | 0     | 1     | 1    | (17) |
| (18) |                                                                                  | PrecStd     | 127    | 20    | 23    | 31    | 34    | 38    | 31    | 15    | 20    | 37    | 27    | 23    | 27   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 10.1  | 14.9  | 21.9  | 25.9  | 31.3  | 36.5  | 37.4  | 35.6  | 32.4  | 26.6  | 18.5  | 12.2 | (19) |
| (20) |                                                                                  | 0.4%        | MCWB   | 4.9   | 6.6   | 9.5   | 11.6  | 14.1  | 16.4  | 17.4  | 16.4  | 15.4  | 12.4  | 9.0   | 6.4  | (20) |
| (21) |                                                                                  | 2%          | DB     | 7.7   | 12.0  | 18.7  | 22.5  | 28.6  | 33.9  | 36.2  | 34.0  | 30.9  | 23.2  | 16.2  | 9.3  | (21) |
| (22) |                                                                                  | 2%          | MCWB   | 3.9   | 5.4   | 8.4   | 9.9   | 13.2  | 14.9  | 16.7  | 15.8  | 14.6  | 11.3  | 7.9   | 4.8  | (22) |
| (23) |                                                                                  | 5%          | DB     | 6.0   | 9.6   | 16.3  | 20.2  | 26.2  | 32.1  | 35.0  | 32.9  | 29.0  | 21.0  | 13.9  | 7.1  | (23) |
| (24) |                                                                                  | 5%          | MCWB   | 2.7   | 4.4   | 7.2   | 9.2   | 12.5  | 14.8  | 16.3  | 15.6  | 13.8  | 10.8  | 6.8   | 3.3  | (24) |
| (25) |                                                                                  | 10%         | DB     | 3.9   | 7.5   | 13.8  | 17.9  | 23.8  | 30.0  | 33.6  | 32.0  | 27.3  | 18.7  | 11.6  | 4.9  | (25) |
| (26) |                                                                                  | 10%         | MCWB   | 1.4   | 3.3   | 6.1   | 8.3   | 11.7  | 14.1  | 16.1  | 15.5  | 13.7  | 9.8   | 5.8   | 2.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 6.0   | 7.4   | 10.1  | 12.2  | 15.3  | 17.6  | 19.6  | 18.8  | 17.9  | 13.9  | 10.3  | 6.9  | (27) |
| (28) |                                                                                  | 0.4%        | MCDWB  | 8.4   | 12.8  | 19.6  | 24.2  | 28.0  | 33.4  | 31.4  | 29.5  | 26.7  | 21.7  | 16.3  | 11.1 | (28) |
| (29) |                                                                                  | 2%          | WB     | 4.4   | 6.0   | 8.9   | 10.7  | 14.1  | 16.3  | 18.7  | 18.0  | 16.3  | 12.7  | 8.5   | 5.2  | (29) |
| (30) |                                                                                  | 2%          | MCDWB  | 7.2   | 10.9  | 17.8  | 20.3  | 25.7  | 30.4  | 30.2  | 29.5  | 26.4  | 20.7  | 14.5  | 8.6  | (30) |
| (31) |                                                                                  | 5%          | WB     | 3.0   | 4.9   | 7.6   | 9.7   | 13.1  | 15.5  | 18.0  | 17.3  | 15.4  | 11.6  | 7.4   | 3.8  | (31) |
| (32) |                                                                                  | 5%          | MCDWB  | 5.4   | 8.9   | 15.5  | 18.5  | 24.1  | 29.3  | 29.9  | 29.0  | 25.5  | 19.3  | 13.1  | 6.4  | (32) |
| (33) |                                                                                  | 10%         | WB     | 1.6   | 3.6   | 6.5   | 8.9   | 12.2  | 14.9  | 17.4  | 16.7  | 14.5  | 10.4  | 6.3   | 2.4  | (33) |
| (34) |                                                                                  | 10%         | MCDWB  | 3.8   | 7.0   | 13.1  | 16.8  | 22.5  | 28.0  | 29.7  | 28.5  | 24.2  | 17.9  | 11.1  | 4.9  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 7.2   | 8.2   | 10.2  | 10.9  | 11.8  | 13.7  | 13.9  | 13.4  | 12.9  | 11.2  | 9.3   | 7.2  | (35) |
| (36) |                                                                                  | 5% DB       | MCDDBR | 8.6   | 10.3  | 13.5  | 14.6  | 15.0  | 16.0  | 15.3  | 14.7  | 14.8  | 13.8  | 11.9  | 8.7  | (36) |
| (37) |                                                                                  |             | MCWBR  | 5.7   | 6.0   | 6.7   | 6.8   | 5.8   | 5.7   | 5.4   | 5.5   | 6.0   | 6.7   | 6.6   | 5.7  | (37) |
| (38) |                                                                                  |             | MCDDBR | 8.1   | 9.5   | 12.7  | 13.4  | 13.8  | 14.7  | 13.0  | 12.6  | 12.4  | 12.3  | 10.9  | 8.2  | (38) |
| (39) |                                                                                  | 5% WB       | MCWBR  | 5.5   | 5.7   | 6.5   | 6.4   | 5.6   | 5.6   | 4.8   | 5.0   | 5.3   | 6.0   | 6.4   | 5.5  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.253  | 0.265 | 0.280 | 0.316 | 0.319 | 0.316 | 0.344 | 0.338 | 0.308 | 0.284 | 0.263 | 0.250 | (40) |      |
| (41) |                                                                                  | taud        | 2.467  | 2.503 | 2.547 | 2.440 | 2.483 | 2.530 | 2.470 | 2.480 | 2.551 | 2.602 | 2.590 | 2.515 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 924    | 963   | 981   | 957   | 955   | 955   | 924   | 922   | 936   | 930   | 909   | 902   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 74     | 84    | 91    | 109   | 107   | 103   | 108   | 104   | 90    | 76    | 66    | 65    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.97   | 2.85  | 4.30  | 5.53  | 6.52  | 7.77  | 7.54  | 6.56  | 5.30  | 3.65  | 2.35  | 1.63  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.21   | 0.32  | 0.29  | 0.34  | 0.53  | 0.54  | 0.33  | 0.30  | 0.35  | 0.24  | 0.23  | 0.21  | (45) |      |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |      |
|------|------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |      |
| (46) | Station Only           | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) | (46) |
|      |                        | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |      |
| (47) | Regional (9 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A | (47) |

Nomenclature: See separate page